

HEALTHBRIDGE · HEALTHCARE DATA PLATFORM

From paper to FHIR R4

Photos and PDFs in. Structured, coded, auditable care records out.

FASTAPI · REACT 18 · POSTGRESQL · FHIR R4 · 10 AGENTS

Gonela Abhilash · CPC · registered pharmacist · MSc Public Health, ARU

Independent project · no patient data shown · information taken from the codebase

WHAT IT IS

One platform: ingest the paper, then watch the patient

21	151	24	10	R4
API MODULES	REST ENDPOINTS	CLINICAL SCREENS	CARE AGENTS	FHIR RECORDS

The problem in the code's own words

Patient data arrives as handwritten prescriptions, printed lab reports, WhatsApp photos of bills and scanned discharge summaries. None of it reaches the doctor, the nurse or the coordinator who needs it.

What drives the build

- One record per patient, not six documents
- Every clinic is a tenant — isolation is in the query layer, not a promise
- Nothing is filed until a human confirms it

Built on

FastAPI 0.115 · Python 3.12 · SQLAlchemy 2 async · PostgreSQL (asyncpg) with SQLite for local dev · Alembic migrations · JWT scoped per tenant · React 18.3 + TypeScript 5.6 + Vite 6 · Flask dashboard for the agent loop.

Three layers, as implemented

- **Ingestion & API** — document services OCR a photo or PDF, convert it, then serve it over a FHIR R4 API
- **Care orchestration** — ten agents on a loop against the same database
- **Multi-tenant SaaS** — organisation → users → patients, subscription tiers, role-based access

WHAT IT DOES

The clinic day, module by module

Clinical workflow

- **OPD registration + token queue** — UHID issued, queue pushed over a WebSocket, doctor actions CALL_NEXT / SKIP / RECALL / COMPLETE
- **Vitals** — typed observations with per-type units and reference ranges, latest-per-type for the chart
- **SOAP notes** — ICD-10 search, autosave, version history, finalise, PDF export
- **Prescriptions** — drug search, formulary detail, interaction checks, allergy warnings, dosing guidelines, then dispense in pharmacy
- **Labs, appointments, family care, risk scores** — separate modules, one patient record

Ingestion & interoperability

- Document ingestion service: photo or PDF → extracted fields → FHIR
- FHIR R4 API plus a generic resource search/read endpoint
- Connectors: ABDM, ABHA, Aadhaar eKYC, OpenMRS, FHIR standard
- WhatsApp bridge for patient messaging



Code does the reading and the plumbing. A clinician still signs off every coded fact before it lands in the record.

Ten agents, and the compliance that makes them usable

The ten agents in the orchestrator

patient intake · appointments · medication adherence · follow-up · labs · pharmacy · insurance · risk prediction · family care · voice care

Each runs on a schedule against the same patient record; the Flask dashboard shows status, patient stats and an activity feed.

Who it serves

- **Now:** single clinics and hospitals in India — DPDP 2025 controls, ABDM/ABHA and Aadhaar eKYC connectors
- **Multi-clinic groups:** tenant isolation with Free / Starter / Professional / Enterprise tiers
- **Later, the UK:** the same loop, but it would need DTAC, DSPT, a DPIA and DCB0129 — none of which are claimed here

Compliance, in code

Purpose-based consent, consent withdrawal, scheduled and executed erasure requests, field-level encryption (Fernet), an audit log with its own viewer screen, and an export centre.

Why a practice can adopt it without ripping out its system

The API speaks FHIR R4 and exposes a resource search endpoint, so records can be read back out. Connectors to ABDM and OpenMRS already exist — the point is interoperation, not replacement.

STATUS · AND WHAT WE MEASURE

What is built, what is not, and how you would judge it

Build status, tracked in the repo

A 45-task build loop runs against the codebase; the last report has **33 complete (73%)**. Foundation, compliance and doctor workflow are done; financial and patient-portal phases are in progress. There is no “finished” claim here — the loop is public in the README history.

How a pilot gets judged — three numbers

- **Minutes per incoming document** — before versus after
- **Codes accepted unchanged** — share a clinician keeps as proposed
- **Recalls closed by day 30** — not rolled into the next month

Why work with us

- The coding is the day job — CPC-certified, risk adjustment, real charts
- Everything shown came out of the repository, not a mock-up
- Limits stated up front: no NHS customer, no UK certification yet
- Built to be reviewed — 151 endpoints and 24 screens are auditable



If those three do not move in thirty days, the pilot failed — and we say so first.

One clinic. One inbox. Thirty days.

Then you judge it on three numbers, not on a demo.

Abhilashgonela886@gmail.com

+44 7344 796772

linkedin.com/in/abigonela

See the loop in 30 seconds

Six clickable screens, sample data, no sign-up.

19p1362.github.io/portfolio/healthbridge

Independent project. Not affiliated with the NHS. No DTAC, DSPT or DCB0129 claim is made. Source code is private — a walkthrough is available on request.